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Outline

- What is ML?
 - ML – Why, What and When?
- Types of Machine Learning
 - Supervised learning
 - Unsupervised learning
 - Semi-supervised learning
 - Reinforcement learning
- Designing a Learning System



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ML – Why?

- Big data
 - Large datasets from growth of automation/web.
E.g., Web click data, medical records, biology, engineering
- Applications can't program by hand.
 - E.g., Autonomous helicopter, handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.
- Self-customizing programs
 - E.g., Amazon, Netflix product recommendations
- Understanding human learning (brain, real AI).

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ML – What?

- “Learning is any process by which a system improves performance from experience.” - **Herbert Simon**
- “Learning Problem: A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P**, if its performance on **T**, as measured by **P**, improves with experience **E**.” - **Tom Mitchell (1998)**

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AI – The traditional approach

Traditional Programming



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Challenges with the traditional approach



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Challenges with the traditional approach



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Machine Learning – Our hero?

Machine Learning



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Machine Learning – Our hero?

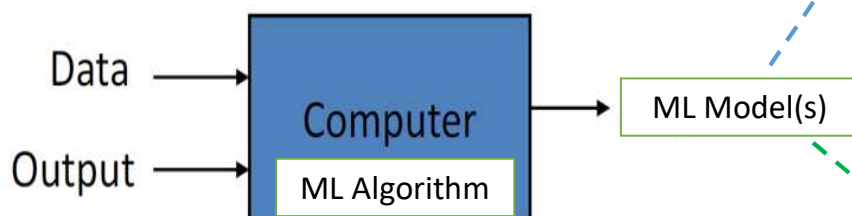
Machine Learning



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Machine Learning – Our hero?

Machine Learning



$$y = f(x_1, x_2, x_3, x_4) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

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So, How does a machine learn?

- A Machine Learning problem



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Leonardo DiCaprio

Ben Affleck

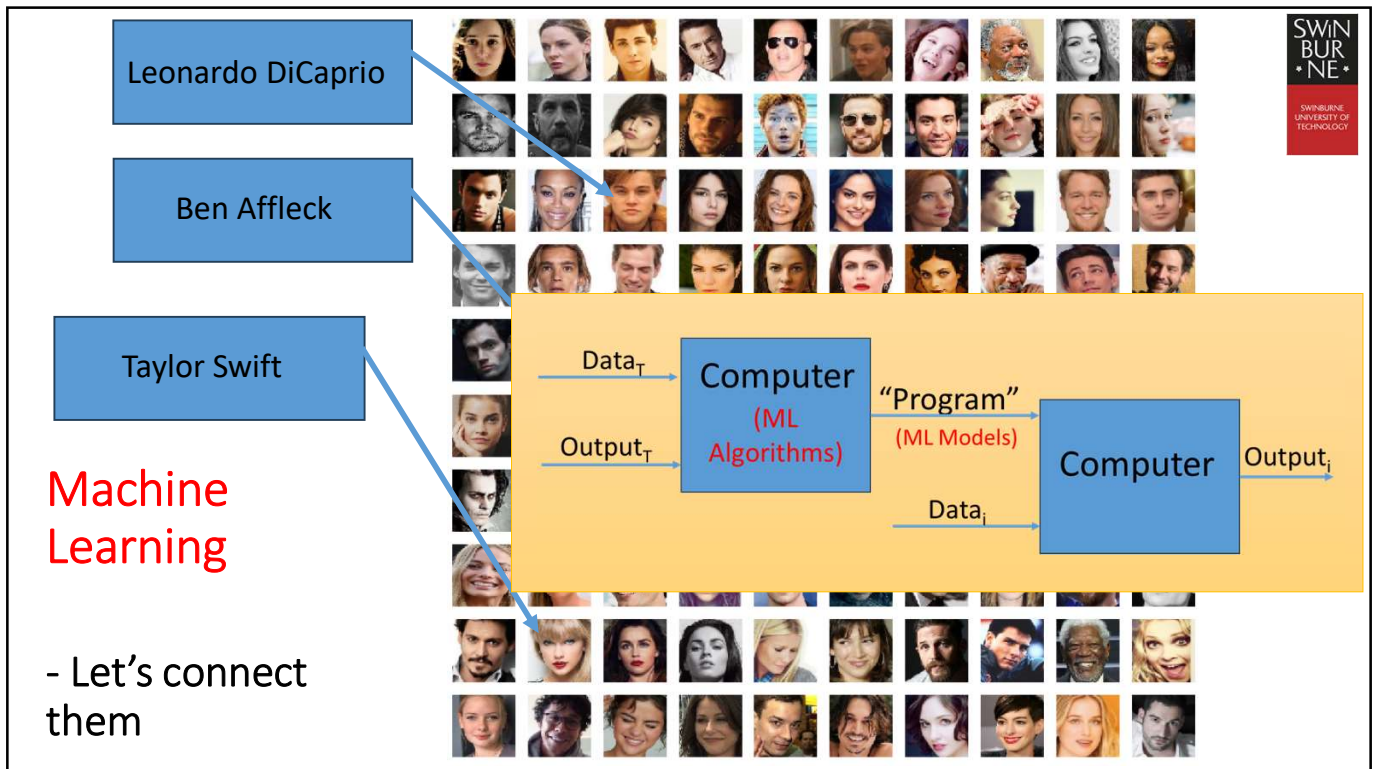
Taylor Swift

So, How does a machine learn?

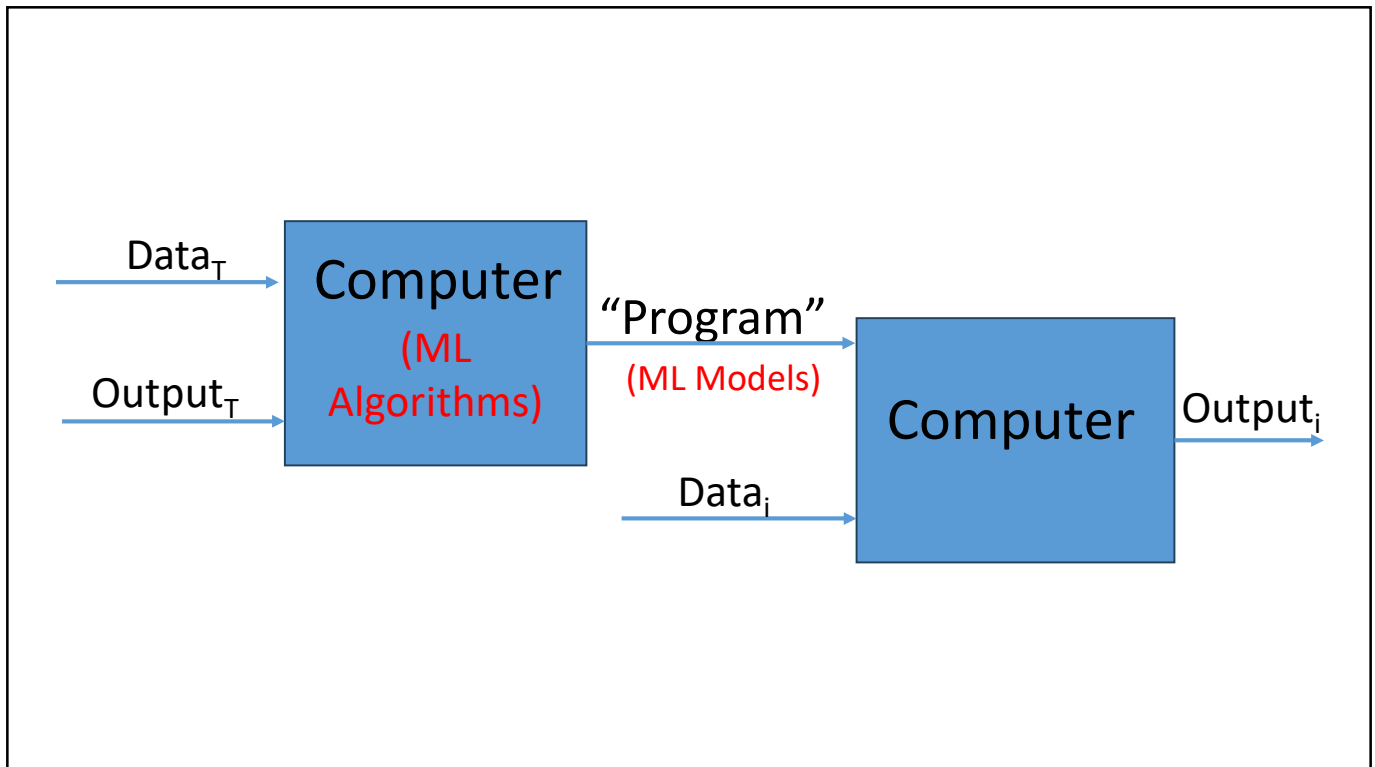
- A Machine Learning problem



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ML vs Traditional programming

Traditional Programming



Machine Learning



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ML – How does Machine learn?

- From a dataset consisting of multiple observations/ examples
- From (direct) experiences



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ML – How does Machine learn?

- From a dataset consisting of multiple observations/examples
- From (direct) experiences



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ML – How does Machine learn?

- From a dataset consisting of multiple observations/examples



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ML – How does Machine learn?

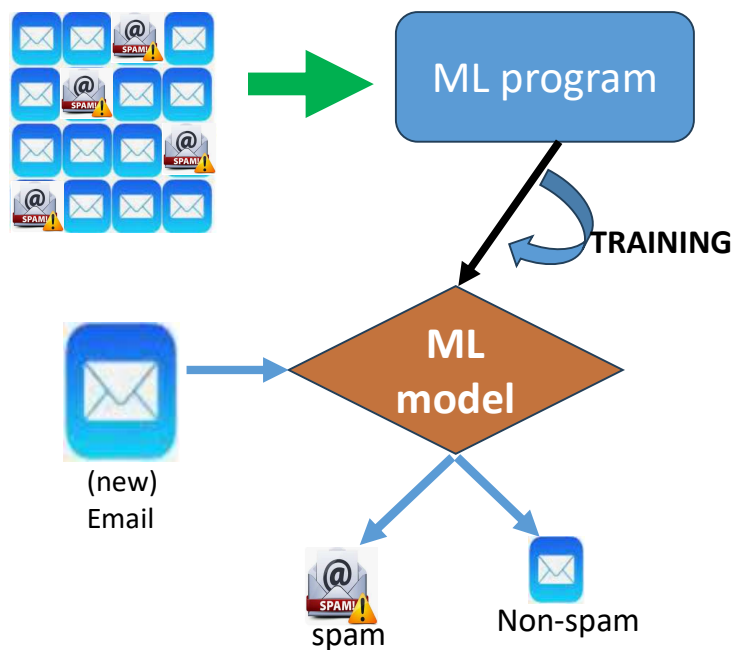
- From a dataset consisting of multiple observations/examples



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ML – How does Machine learn?

- From a dataset consisting of multiple observations/examples



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ML – How does Machine learn?

- ML algorithms and terminologies



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ML – How does Machine learn?

The “play tennis” dataset

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

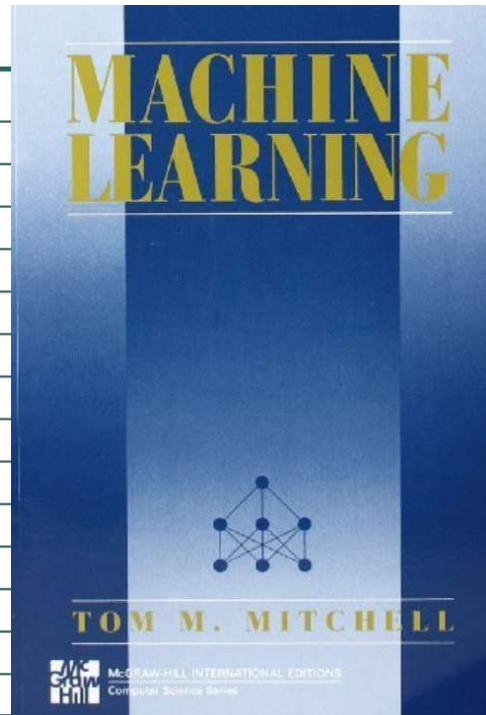
Source: Tom Mitchell, 1997

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ML – How does Machine learn?

The “play tennis” dataset

Outlook	Temperature
Sunny	Hot
Sunny	Hot
Overcast	Hot
Rainy	Mild
Rainy	Cool
Rainy	Cool
Overcast	Cool
Sunny	Mild
Sunny	Cool
Rainy	Mild
Sunny	Mild
Overcast	Mild
Overcast	Hot
Rainy	Mild



Source: Tom Mitchell, 1997

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ML – How does Machine learn?

The “play tennis” dataset

Examples/
Observations

⋮

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Source: Tom Mitchell, 1997

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ML – How does
Machine learn?

Features/
Attributes

Target

The “play tennis” dataset

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

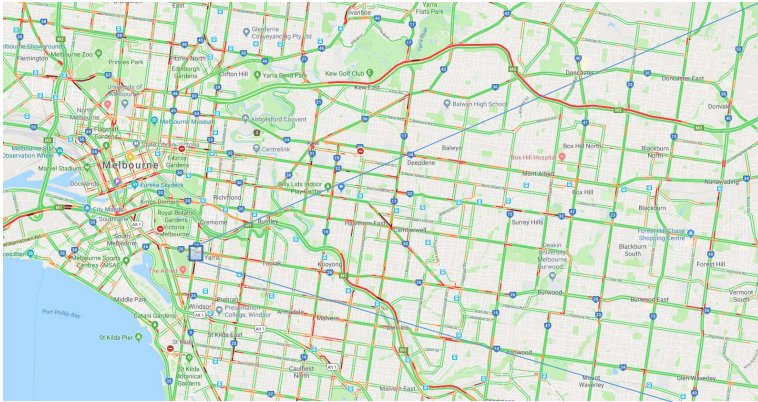
Source: Tom Mitchell, 1997

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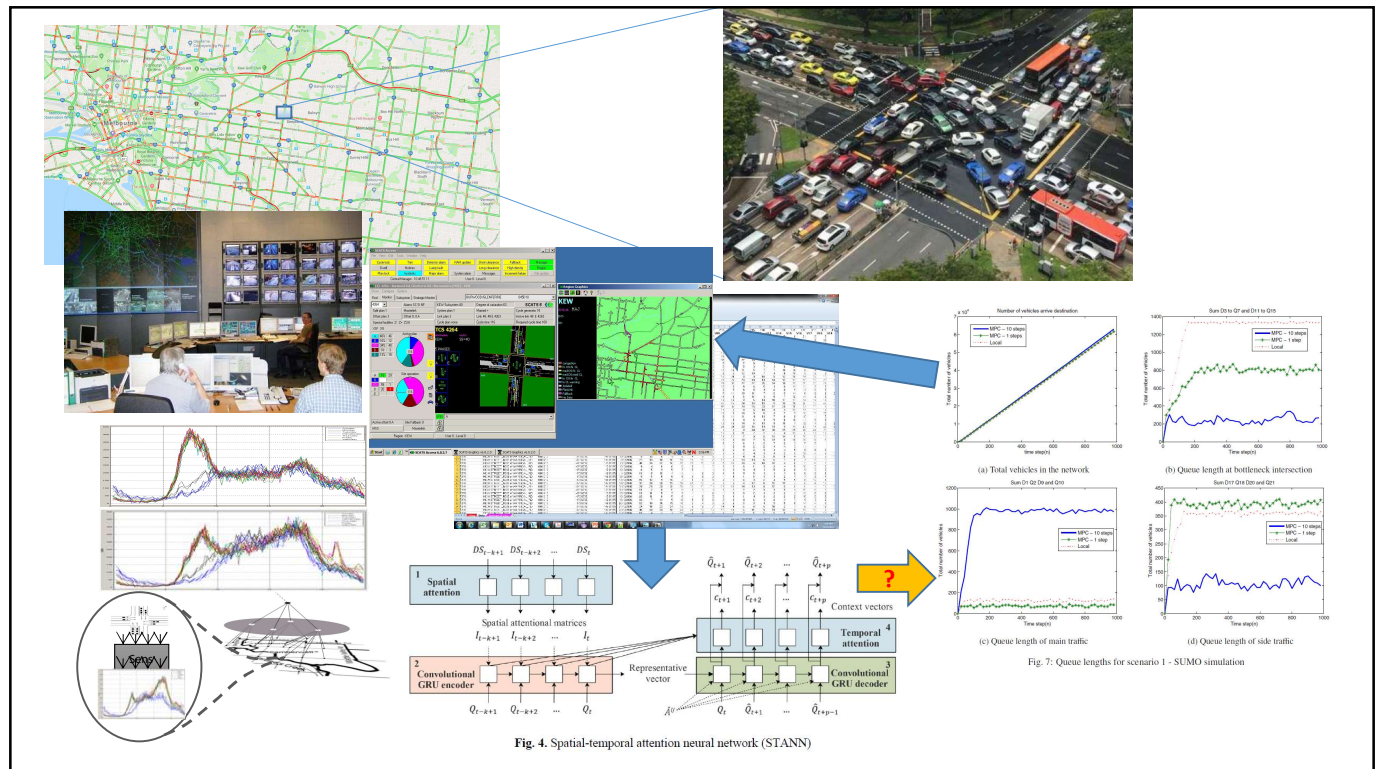
State of the Art Applications of Machine Learning

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AI & ML @Swinburne



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Types of Machine Learning

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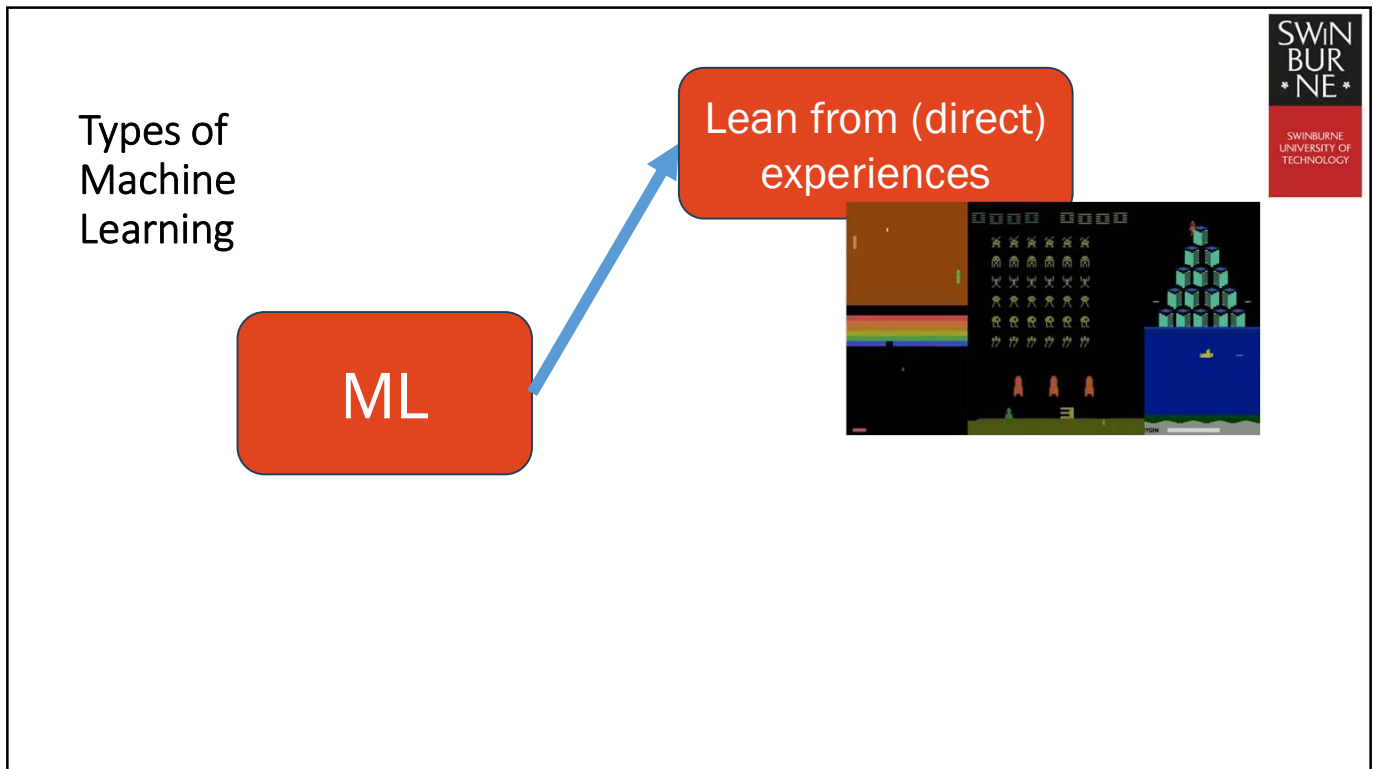
Types of
Machine
Learning

ML

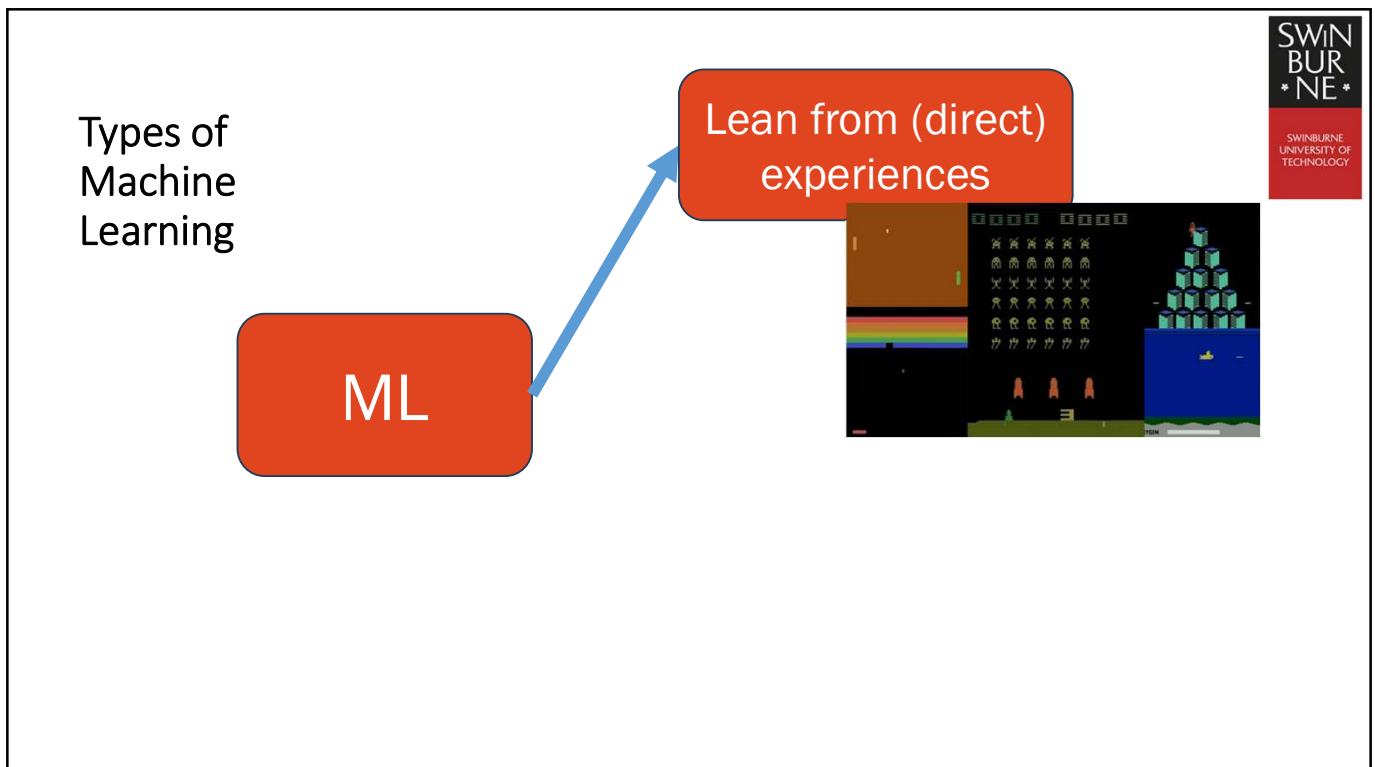
Learn from (direct)
experiences



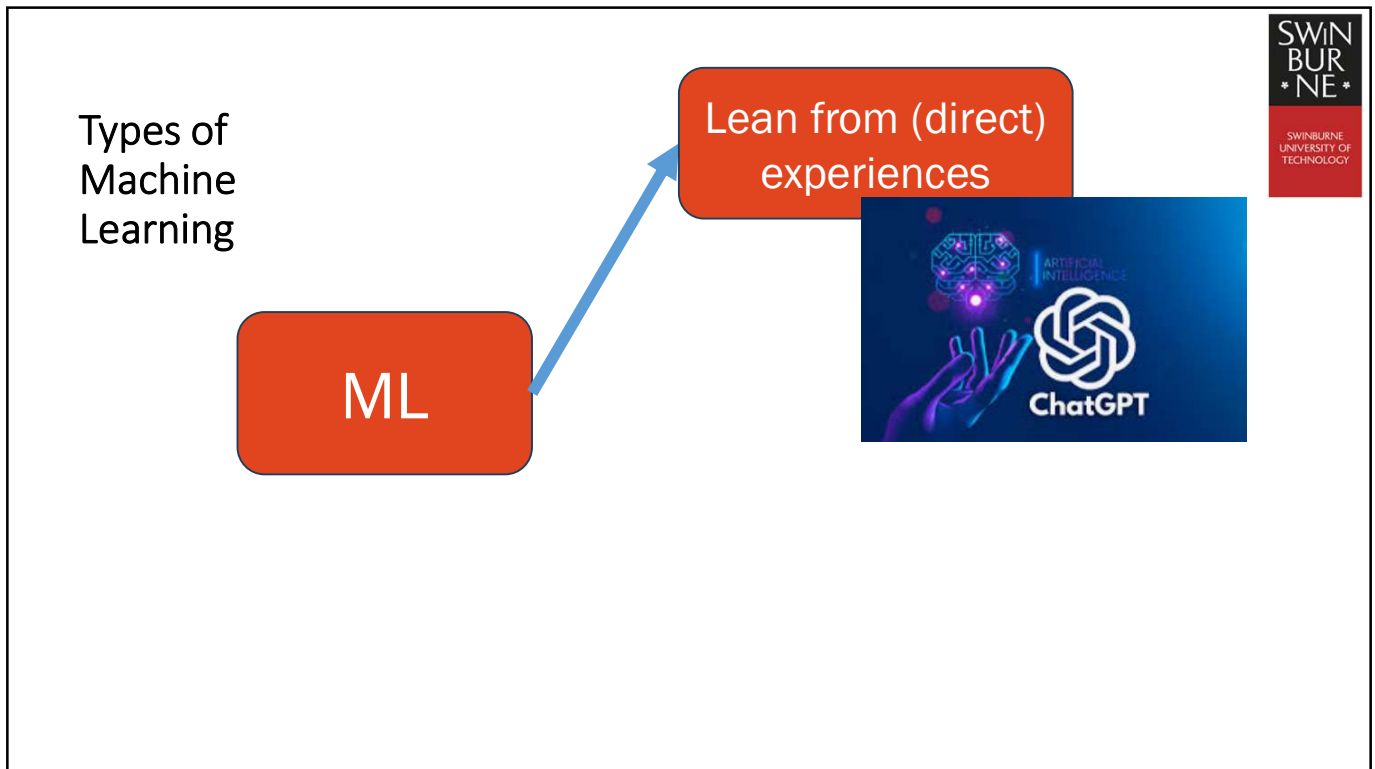
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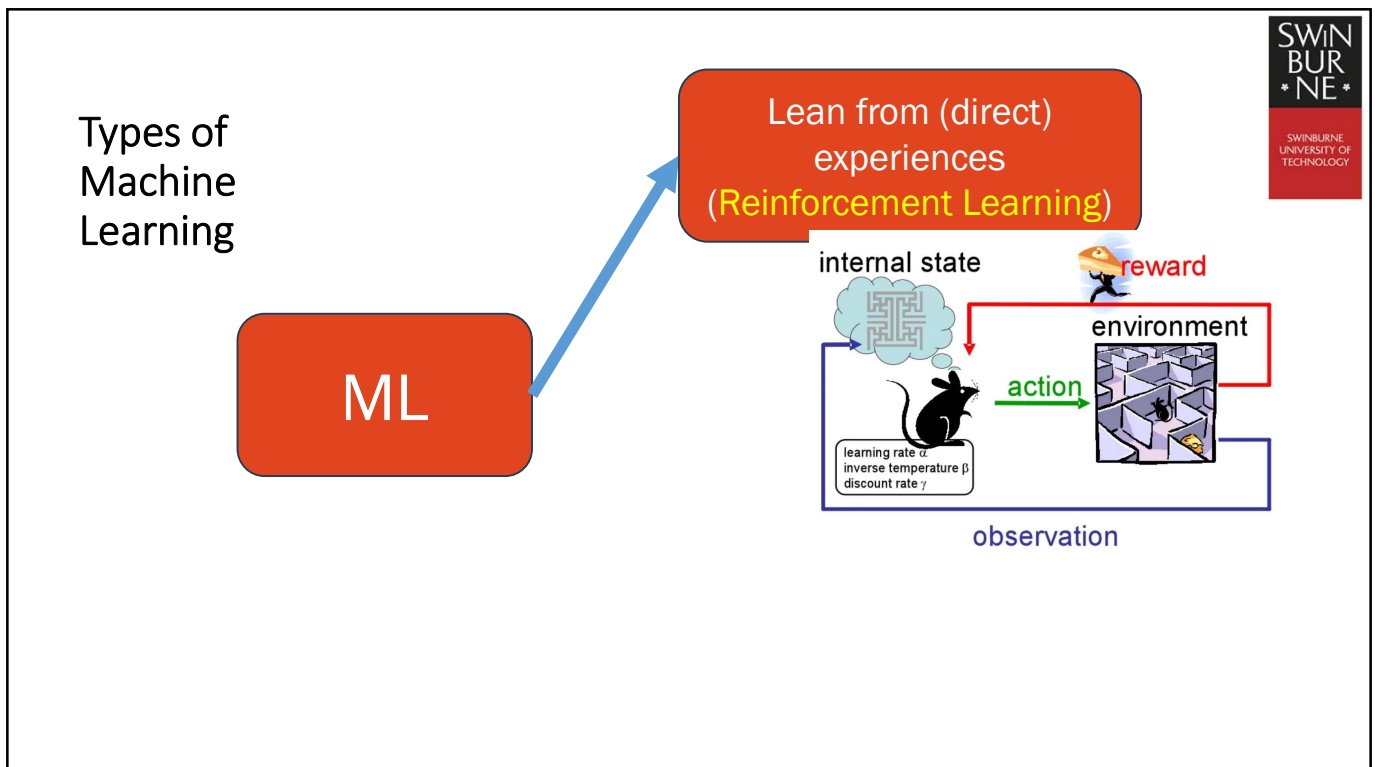
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Reinforcement Learning

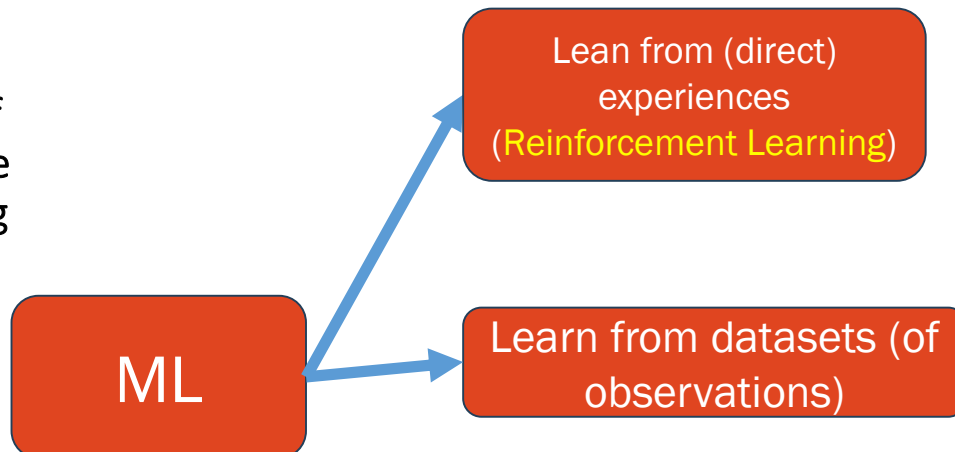
Learn policy from user demonstrations



Stanford Autonomous Helicopter
<http://heli.stanford.edu/>

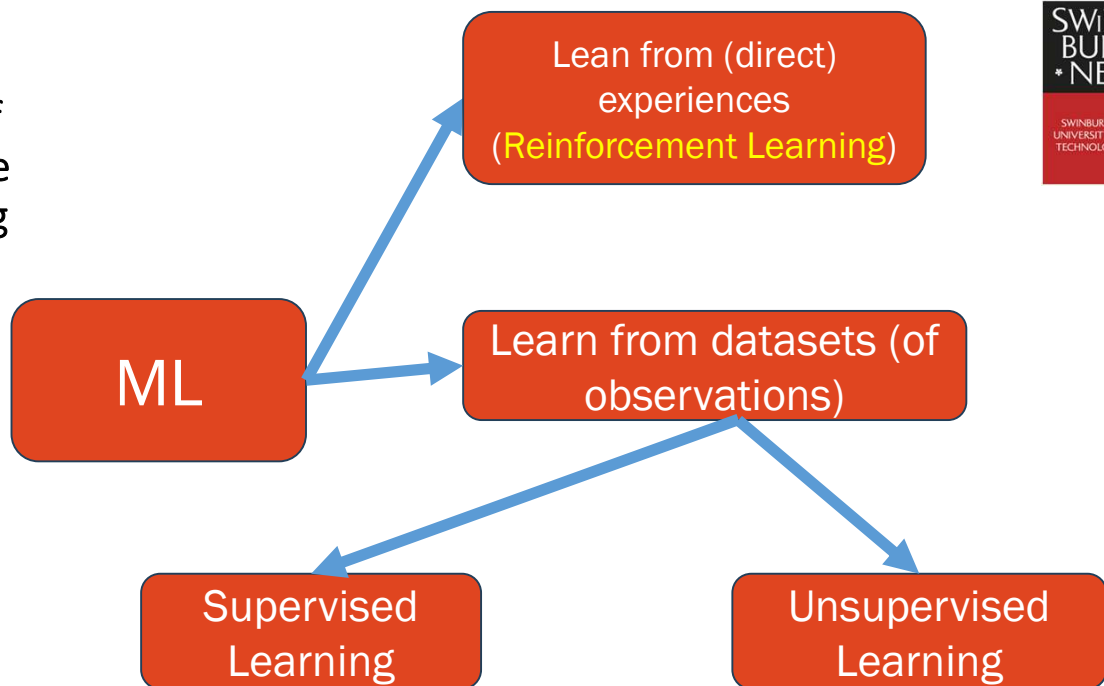
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Types of
Machine
Learning



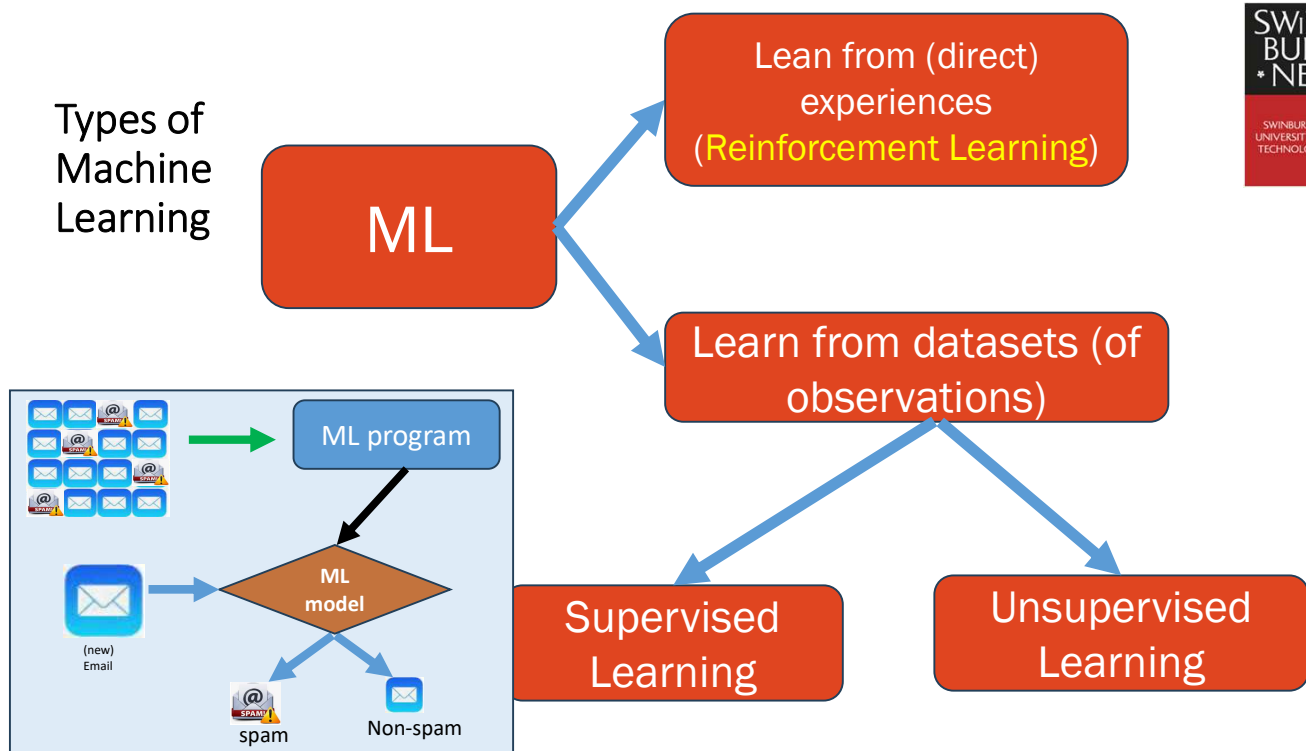
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Types of Machine Learning

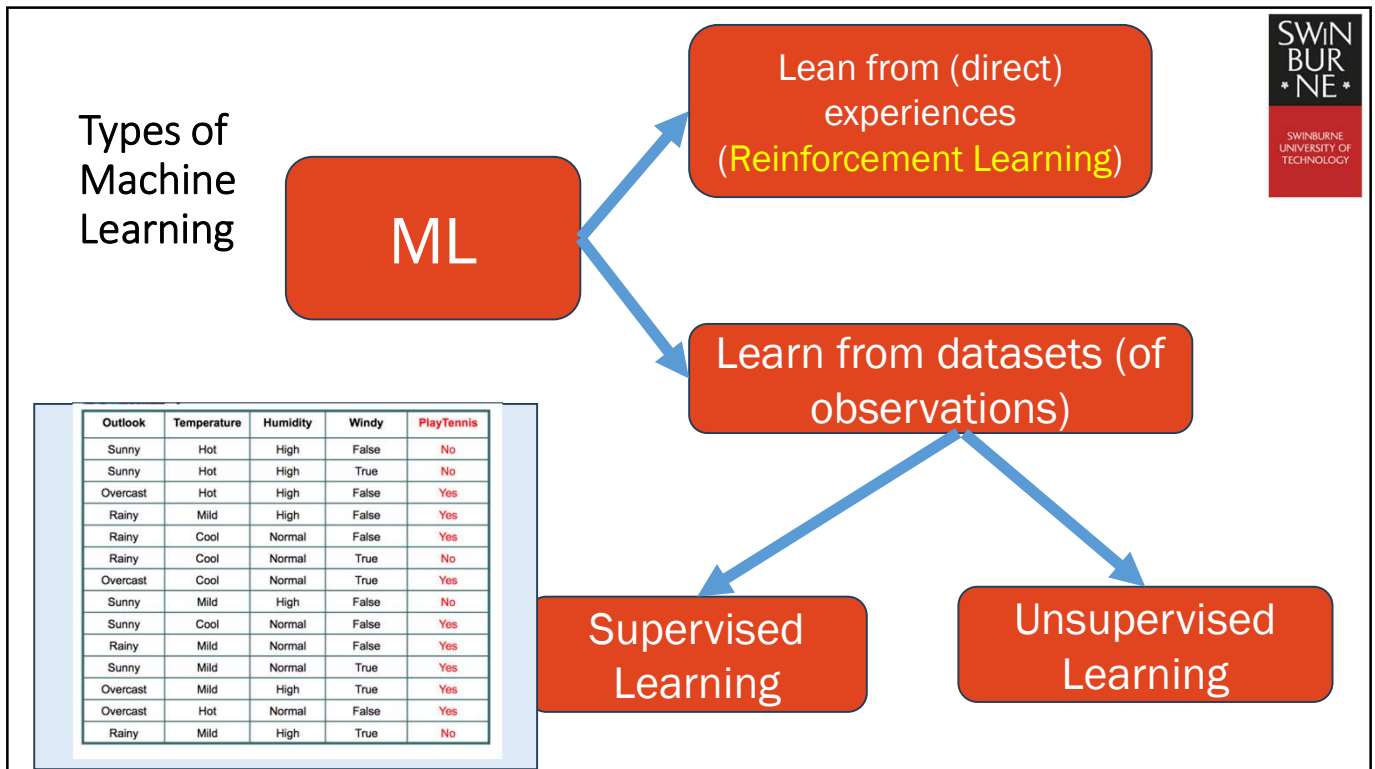


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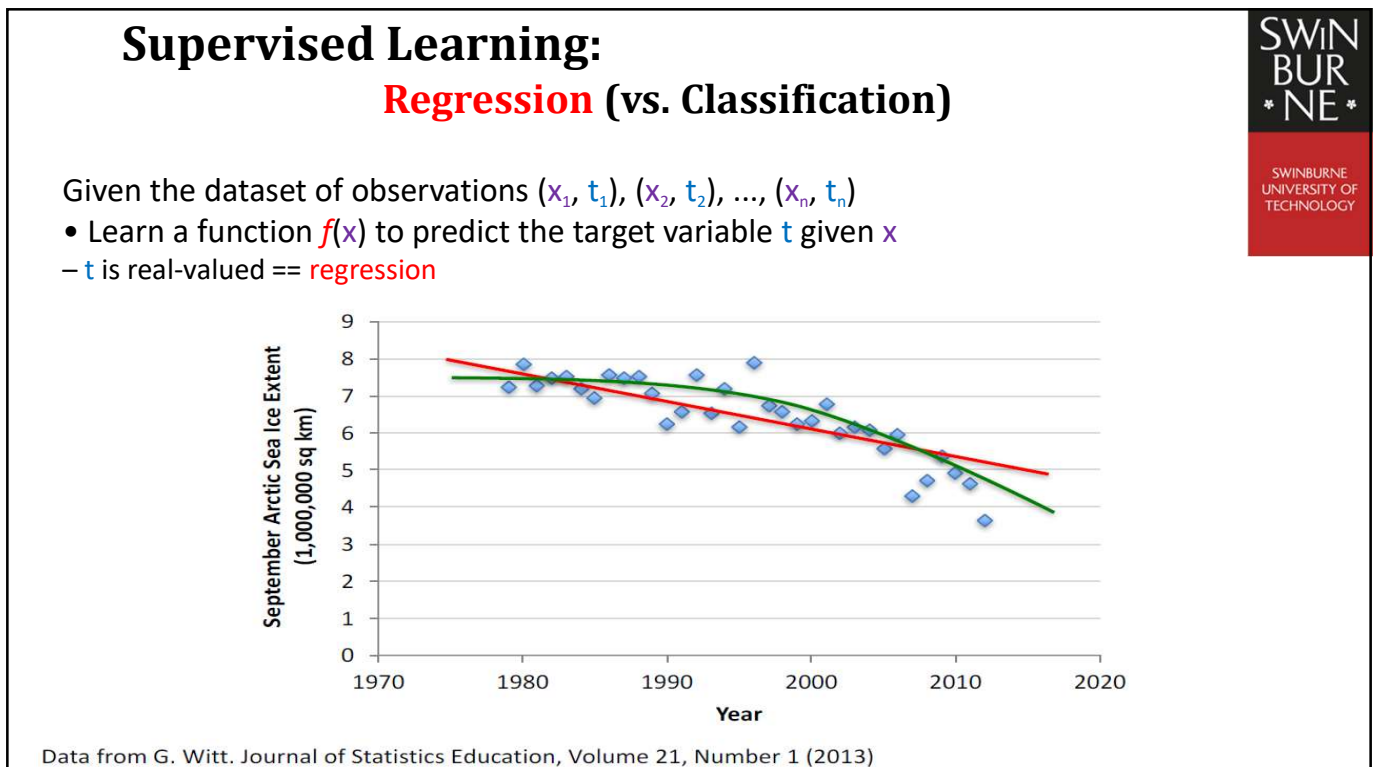
Types of Machine Learning



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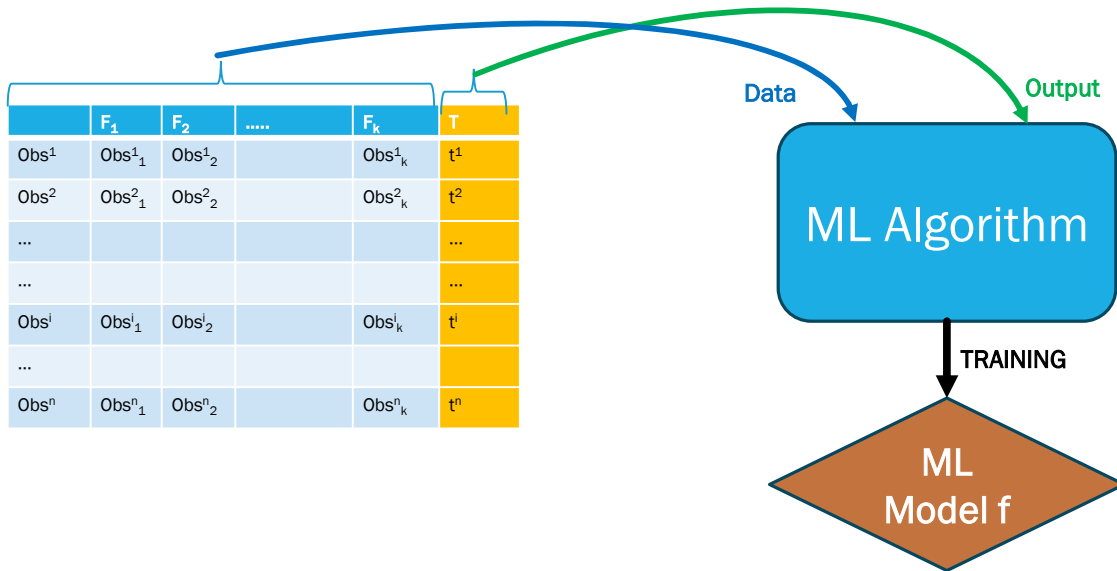


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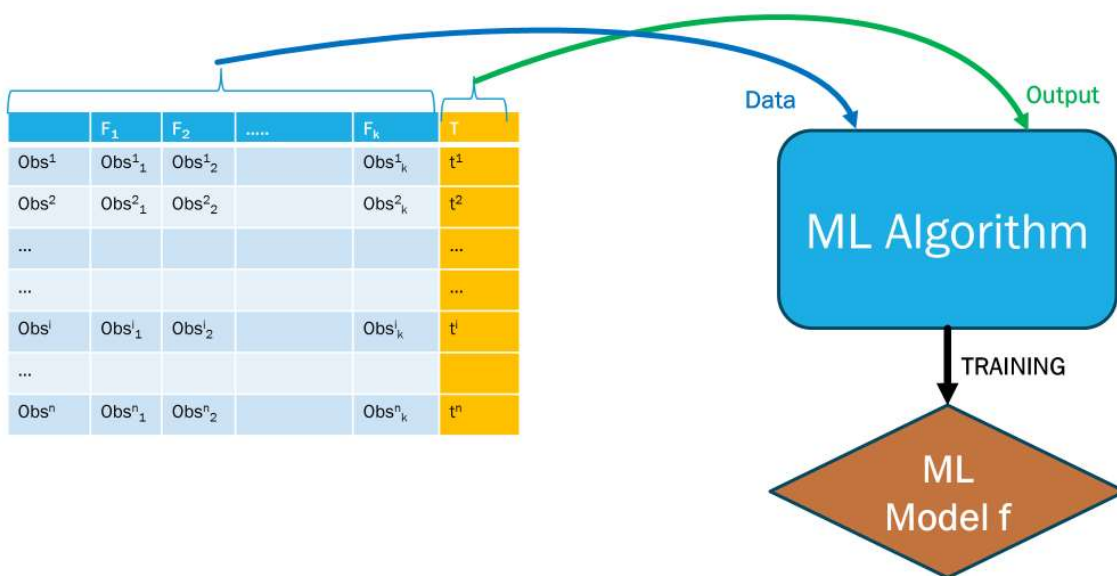
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SUPERVISED LEARNING



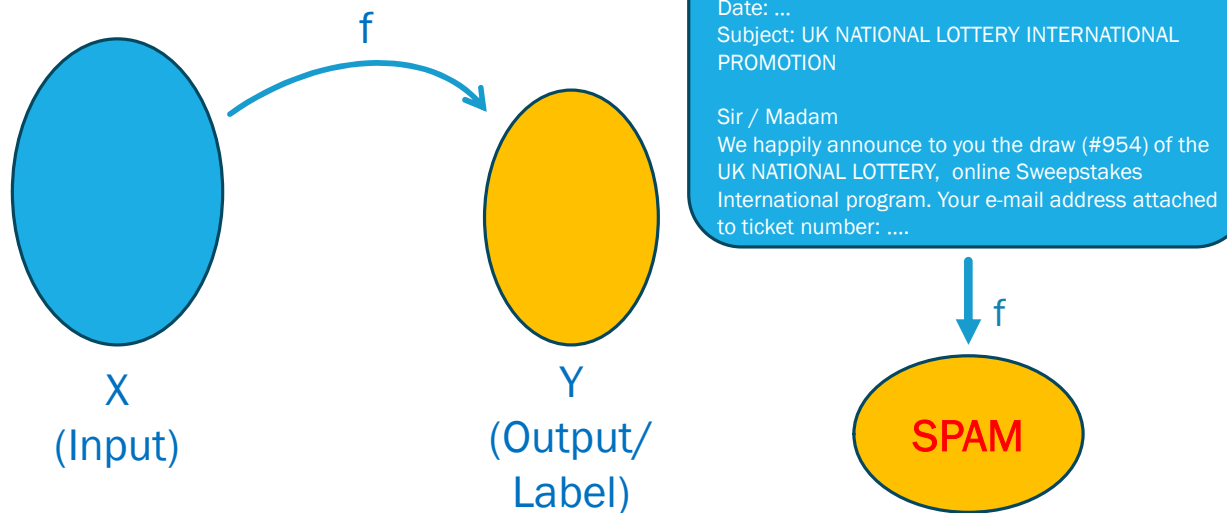
43

SUPERVISED LEARNING



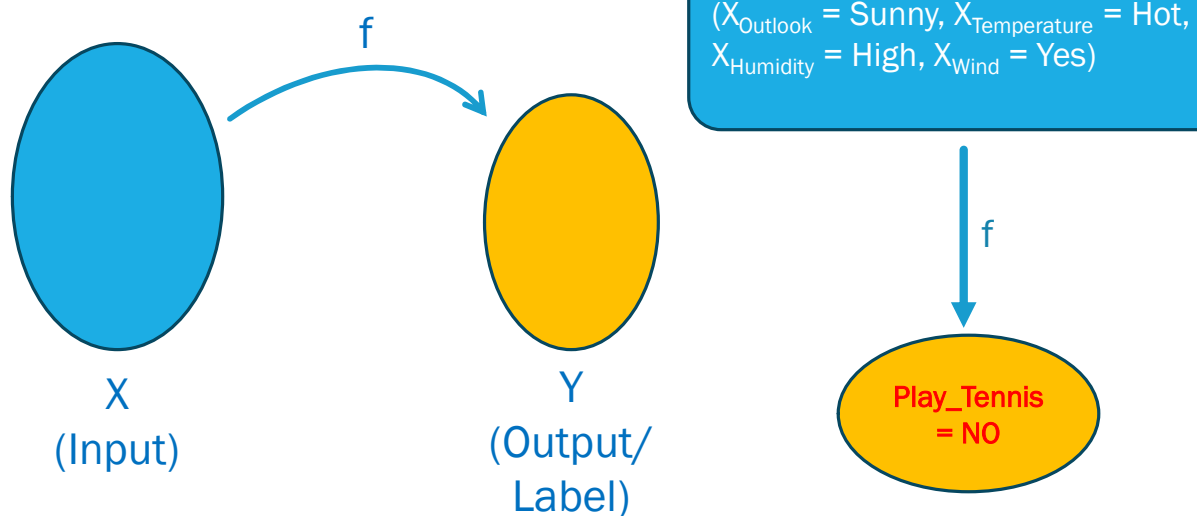
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SUPERVISED LEARNING



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SUPERVISED LEARNING



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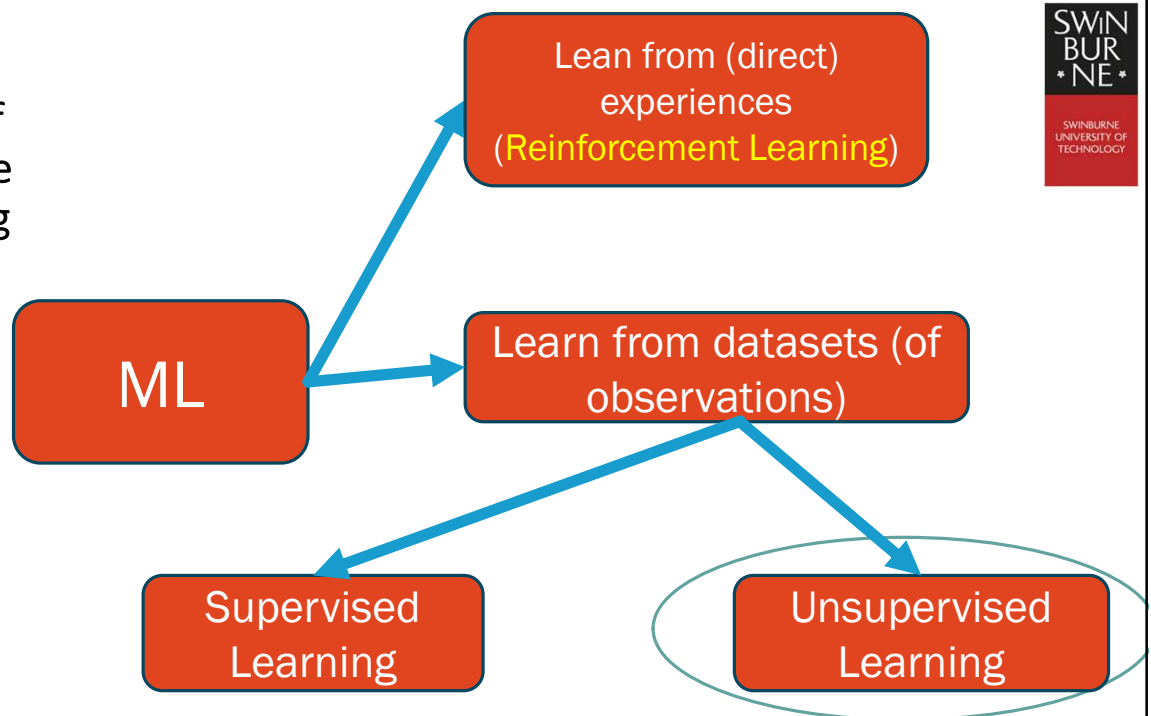
SUPERVISED LEARNING

Input (X)	Output (Y)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Spanish	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	self-driving car
image of phone	defect? (0/1)	visual inspection

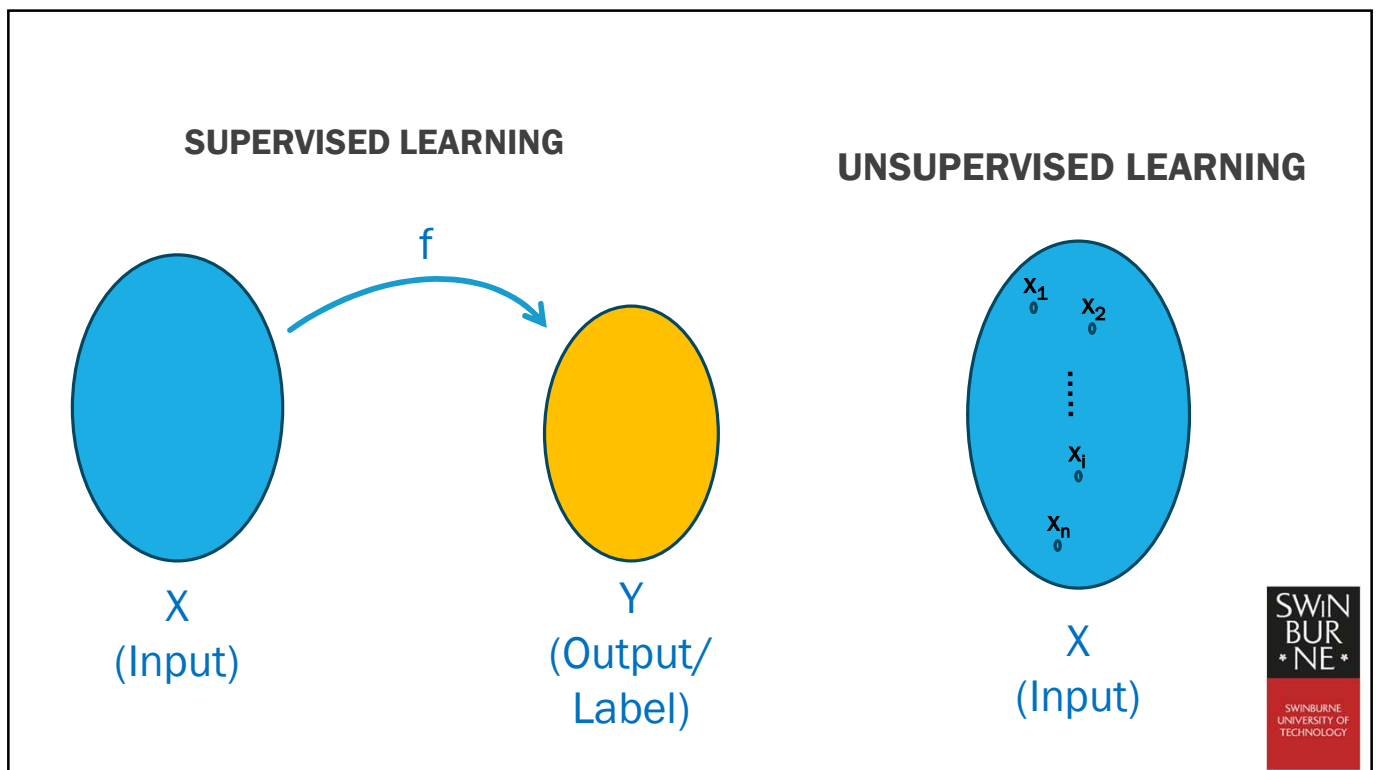
Source: Andrew Ng/Stanford Online

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Types of Machine Learning



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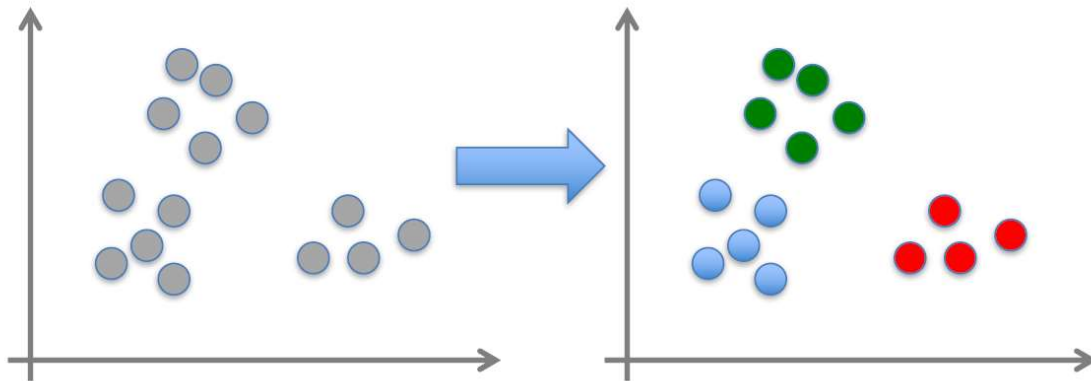


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Unsupervised Learning

Given $Obs_1, Obs_2, \dots, Obs_n$ (without labels)

- Output hidden structure behind the Obs 's
- E.g., clustering



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Unsupervised Learning

Google News

Search for topics, locations & sources

Home For you Following News Showcase Australia World Local Business Technology Entertainment Sports Science Health

Headlines Follow

NEWS
Ukraine agrees to terms of critical minerals deal with US, sources say
5 hours ago

Sydney Morning Herald
"Very big deal": Zelensky set to meet Trump after reaching agreement
5 hours ago • By Michael Koziol

The Conversation
Why Trump really wants Ukraine's minerals – China has put theirs off limits
10 hours ago • By Dafydd Townley

The Australian
Zelensky to sign minerals deal with Trump in US visit
1 hour ago • By Ian Lovett and Alex Ward

Full coverage

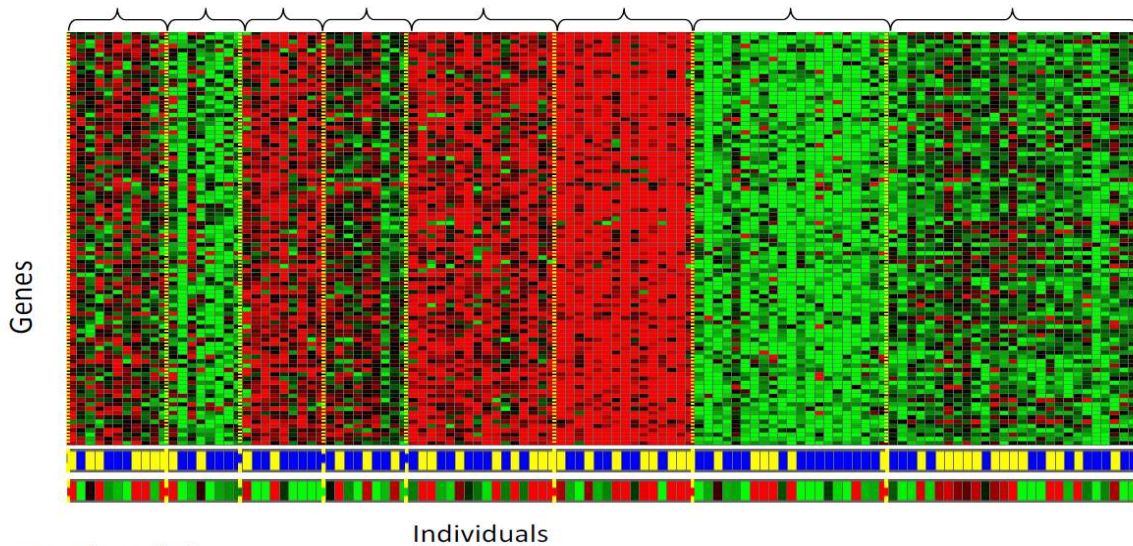
ABC News
Federal politics live: Dutton demands answers from PM over Chinese warship live-fire drill
2 hours ago • By Courtney Gould

E.g., <https://news.google.com> uses clustering

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Unsupervised Learning

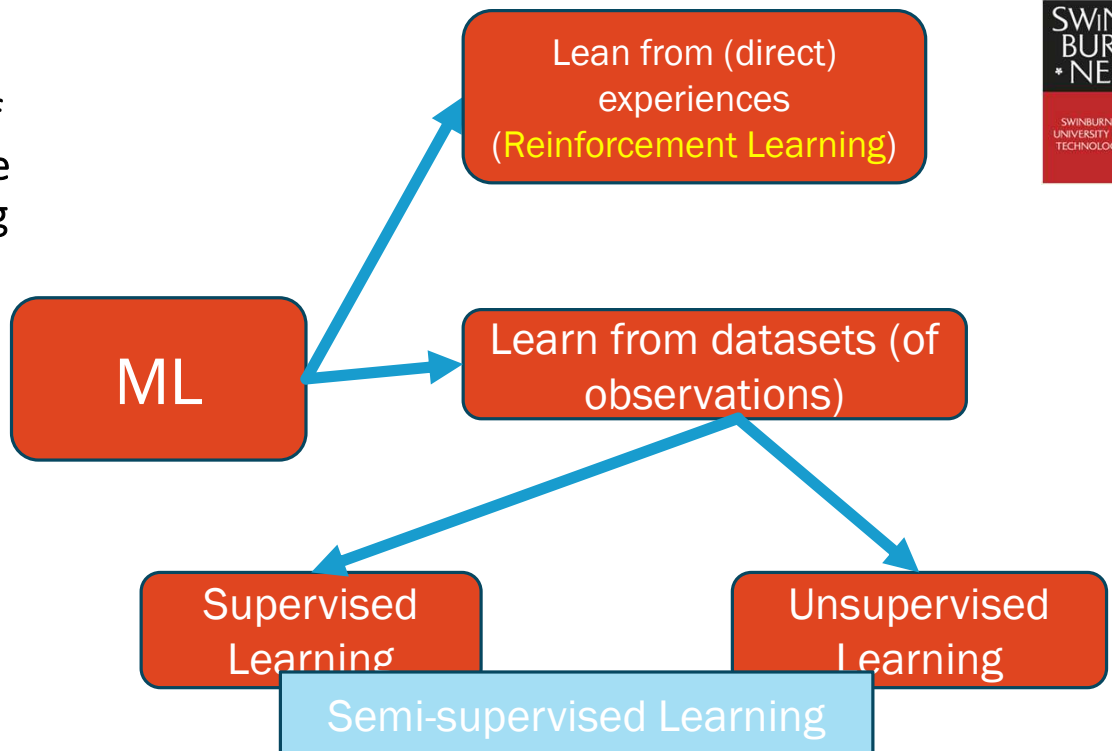
Genomics application: group individuals by genetic



[Source: Daphne Koller]

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Types of
Machine
Learning



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Inductive Learning (Science)

- Simplest form: learn a function from examples

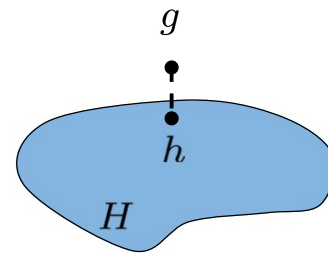
- A target function: g
- Examples: input-output pairs $(x, g(x))$
- E.g. x is an email and $g(x)$ is spam / ham
- E.g. x is a house and $g(x)$ is its selling price

- Problem:

- Given a hypothesis space H
- Given a training set of examples $\mathcal{X}_i$
- Find a hypothesis $h(x)$ such that $h \sim g$

- Includes:

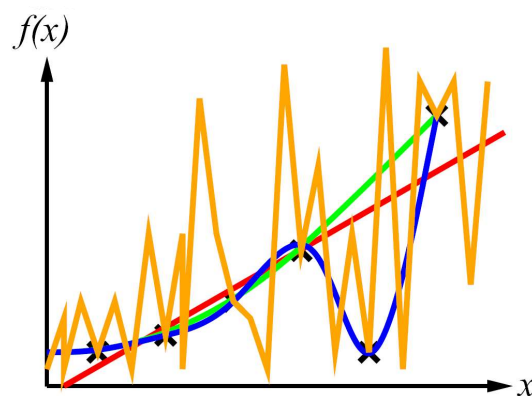
- Classification (outputs = class labels)
- Regression (outputs = real numbers)



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Inductive Learning

- Curve fitting (regression, function approximation):



- Consistency vs. simplicity
- Ockham's razor

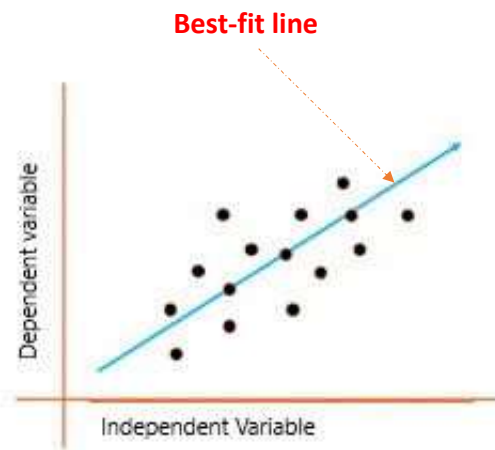
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Supervised Learning with Linear Regression

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Linear Regression (LR)

- A statistical regression method used for predictive analysis
- LR shows the linear relationship between the independent (predictor) variable i.e. X-axis and the dependent (output) variable i.e. Y-axis



Source:

<https://www.analyticsvidhya.com/blog/2021/10/everything-you-need-to-know-about-linear-regression/>

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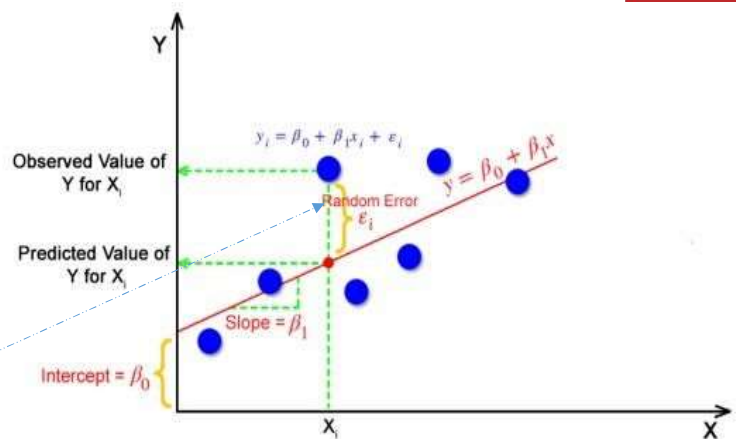
Linear Regression (LR)

- How to compute the **best-fit line**?

$y = h_{\beta}(x) = \beta_0 + \beta_1 x$, where:

- y = Dependent variable,
- x = Independent variable,
- β_0 = constant/Intercept,
- β_1 = Slope.

For each **training example** (x_i, y_i) , we can compute the **random error** (or, **residual**) $\epsilon_i = Y_{\text{predicted}} - Y_i$
 where $Y_{\text{predicted}} = h_{\beta}(x_i) = \beta_0 + \beta_1 x_i$



Source:

<https://www.analyticsvidhya.com/blog/2021/10/everything-you-need-to-know-about-linear-regression/>

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Linear Regression (LR) - Best-fit Line

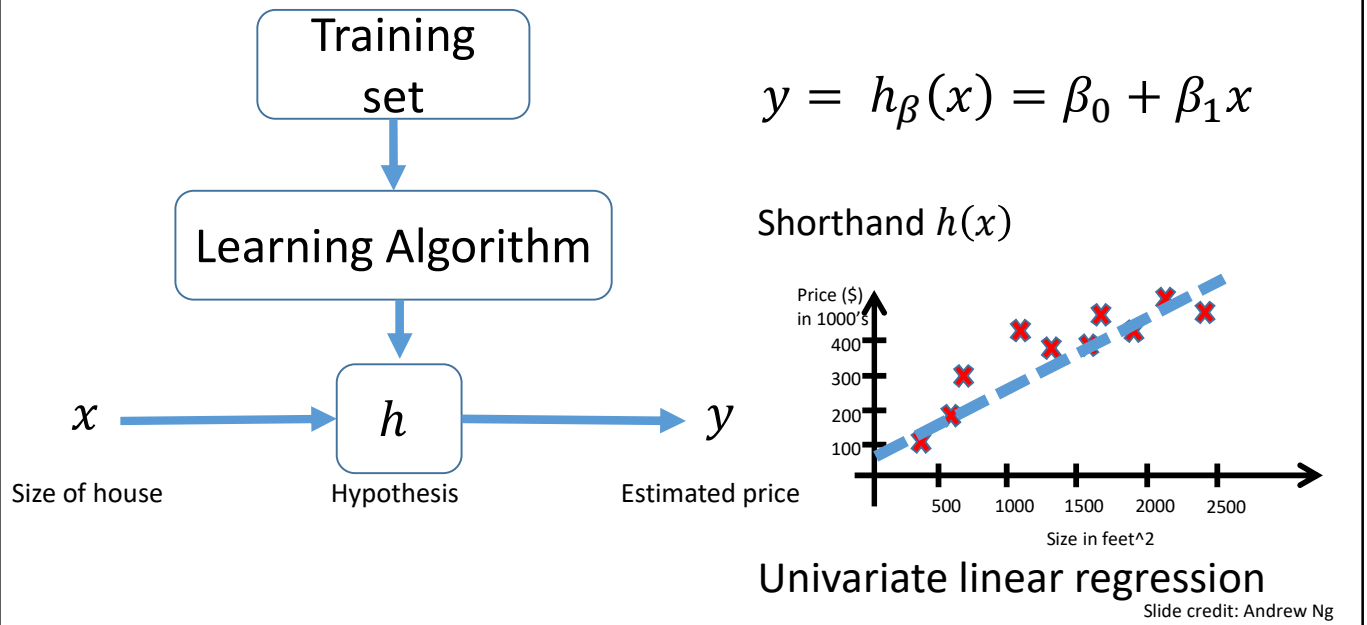
- What is the best fit line, i.e., what are the best values for β_0 and β_1 ?
- Cost Function:**
 - Measuring the error a given line $y = h_{\beta}(x) = \beta_0 + \beta_1 x$ has with respect to a set of training examples $\{(x_1, y_1), \dots, (x_n, y_n)\}$
 - For instance, the **Mean Squared Error (MSE)** cost function:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

- The line $y = h_{\beta}(x) = \beta_0 + \beta_1 x$ is "best fit" if MSE is minimised

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Model representation



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- **Hypothesis:** $h_{\beta}(x) = \beta_0 + \beta_1 x$
- **Parameters:** β_0, β_1
- **Cost function:** $J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i)^2$
- **Goal:** minimize $J(\beta_0, \beta_1)$
 β_0, β_1

Slide credit: Andrew Ng

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Gradient descent

Have some function $J(\beta_0, \beta_1)$

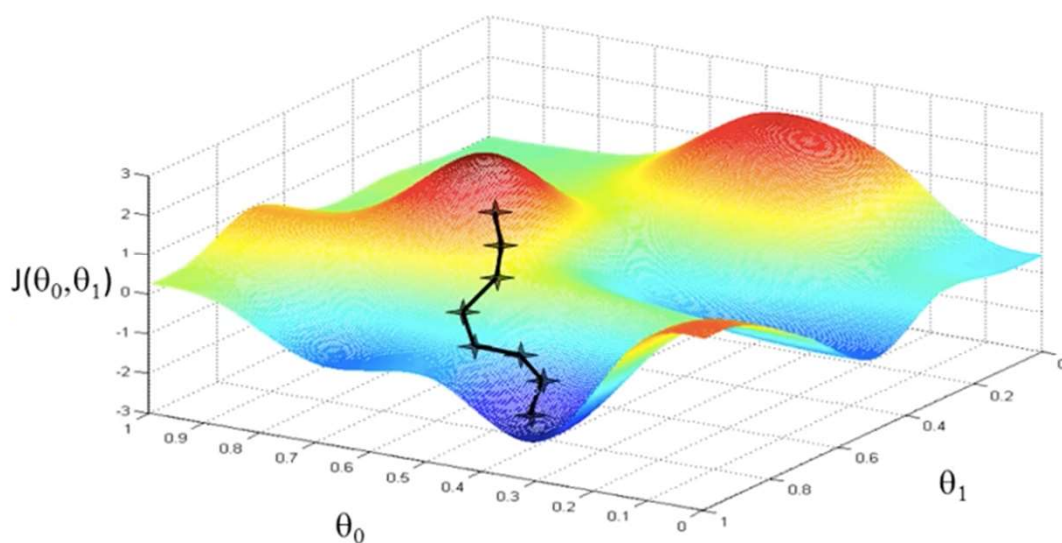
Want $\underset{\beta_0, \beta_1}{\operatorname{argmin}} J(\beta_0, \beta_1)$

Outline:

- Start with some β_0, β_1
- Keep changing β_0, β_1 to reduce $J(\beta_0, \beta_1)$ until we hopefully end up at minimum

Slide credit: Andrew Ng

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Slide credit: Andrew Ng

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Gradient descent

Repeat until convergence {

$$\beta_j := \beta_j - \alpha \frac{\partial}{\partial \beta_j} J(\beta_0, \beta_1) \quad (\text{for } j = 0 \text{ and } j = 1)$$

}

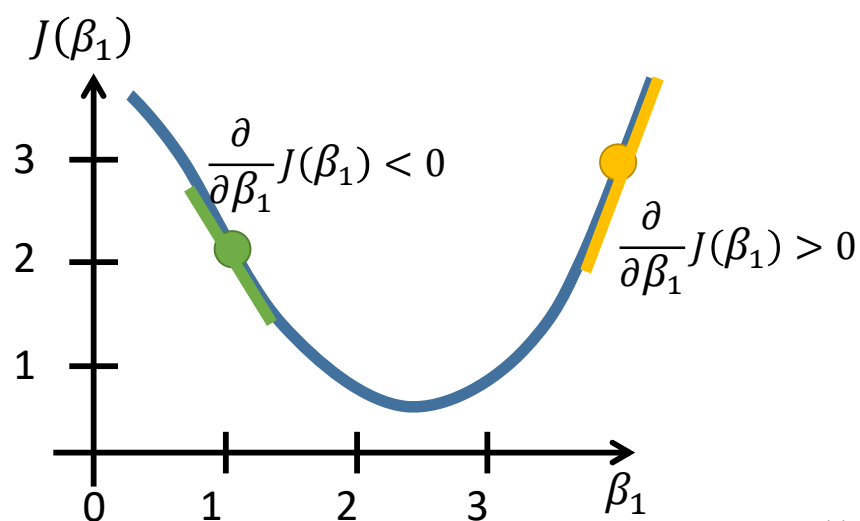
α : Learning rate (step size)

$\frac{\partial}{\partial \beta_j} J(\beta_0, \beta_1)$: derivative (rate of change)

Slide credit: Andrew Ng

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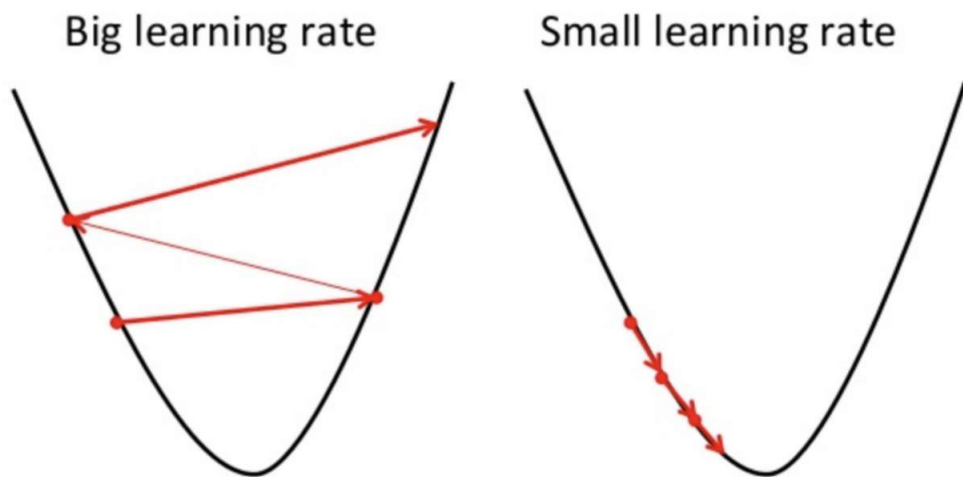
$$\beta_1 := \beta_1 - \alpha \frac{\partial}{\partial \beta_1} J(\beta_1)$$



Slide credit: Andrew Ng

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Learning rate



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Gradient descent for linear regression

Repeat until convergence{

$$\beta_j := \beta_j - \alpha \frac{\partial}{\partial \beta_j} J(\beta_0, \beta_1) \quad (\text{for } j = 0 \text{ and } j = 1)$$

}

- Linear regression model

$$h_{\beta}(x) = \beta_0 + \beta_1 x$$

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i)^2$$

Slide credit: Andrew Ng

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Computing partial derivative

- $$\frac{\partial}{\partial \beta_j} J(\beta_0, \beta_1) = \frac{\partial}{\partial \beta_j} \frac{1}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i)^2$$

$$= \frac{\partial}{\partial \beta_j} \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 x_i - y_i)^2$$
- $$j = 0: \frac{\partial}{\partial \beta_0} J(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i)$$
- $$j = 1: \frac{\partial}{\partial \beta_1} J(\beta_0, \beta_1) = \frac{2}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i) x_i$$

Slide credit: Andrew Ng

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Gradient descent for linear regression

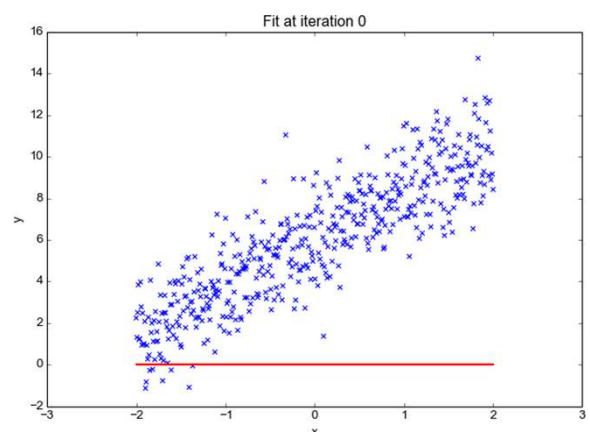
Repeat until convergence {

$$\beta_0 := \beta_0 - \alpha \frac{2}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i)$$

$$\beta_1 := \beta_1 - \alpha \frac{2}{n} \sum_{i=1}^n (h_{\beta}(x_i) - y_i) x_i$$

}

Update β_0 and β_1 simultaneously



Slide credit: Andrew Ng

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Regression: How to

- So, do I have to write a program to calculate the “right” values of **a** and **b** to “minimize” the errors?
- And, how about multi-dimensional datasets?
- Short answer: **Just use an existing library!**
 - E.g., **scikit-learn** (see: <https://scikit-learn.org/stable/>)
 - E.g., **tensorflow** (see: <https://www.tensorflow.org/>)

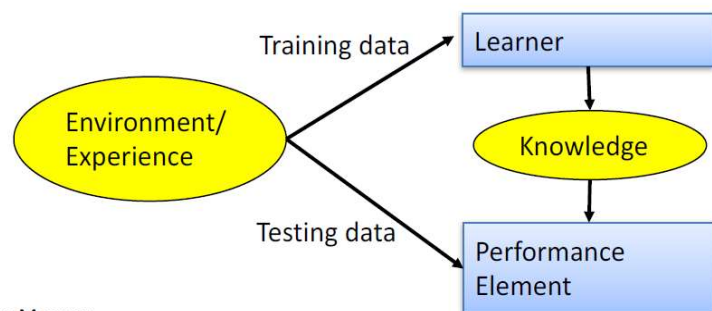
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Designing a Learning System

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Designing a Learning System

- Choose the training experience
- Choose exactly what is to be learned – i.e. the **target function**
- Choose how to **represent** the target function
- Choose a **learning algorithm** to infer the target function from the experience



Based on slide by Ray Mooney

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ML in a Nutshell

- Tens of thousands of machine learning algorithms
 - Hundreds new every year
- Every ML algorithm has three components:
 - **Representation**
 - **Optimization**
 - **Evaluation**

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Various Function Representations



- Numerical functions
 - Linear regression
 - Neural networks
 - Support vector machines
- Symbolic functions
 - Decision trees
 - Rules in propositional logic
 - Rules in first-order predicate logic
- Instance-based functions
 - Nearest-neighbor
 - Case-based
- Probabilistic Graphical Models
 - Naïve Bayes
 - Bayesian networks
 - Hidden-Markov Models (HMMs)
 - Probabilistic Context Free Grammars (PCFGs)
 - Markov networks

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Various Search/Optimization Algorithms



- Gradient descent
 - Perceptron
 - Backpropagation
- Dynamic Programming
 - HMM Learning
 - PCFG Learning
- Divide and Conquer
 - Decision tree induction
 - Rule learning
- Evolutionary Computation
 - Genetic Algorithms (GAs)
 - Genetic Programming (GP)
 - Neuro-evolution

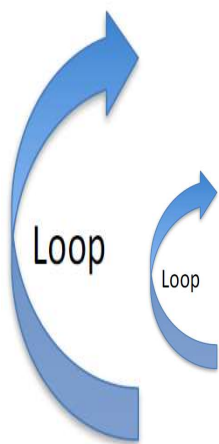
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Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- etc.

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ML in Practice



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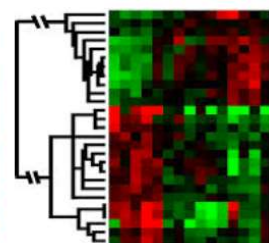
Summary

- Learning can be viewed as using **direct or indirect experience** to approximate a chosen **target function**.
- Function approximation can be viewed as a **search** through a space of **hypotheses** (representations of functions) for one that best fits a set of training data.
- Different learning methods assume different hypothesis spaces (representation languages) and/or employ different search techniques.

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ML – When to Use Machine Learning??

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



Learning isn't always useful:

- There is no need to "learn" to calculate payroll

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A classic example of a task that requires machine learning:
It is very hard to say what makes a 2



Slide credit: Geoffrey Hinton

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ML – Examples

- **Recognizing patterns:**
 - Facial identities or facial expressions
 - Handwritten or spoken words
 - Medical images
- **Generating patterns:**
 - Generating images or motion sequences
- **Recognizing anomalies:**
 - Unusual credit card transactions
 - Unusual patterns of sensor readings in a nuclear power plant
- **Prediction:**
 - Future stock prices or currency exchange rates

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ML for a practitioner



- If you:
 - Are a beginner
 - Need quick results
 - Dataset is simple (small and well-structured, e.g. a CSV file)
- Then consider Weka (<https://www.cs.waikato.ac.nz/ml/weka/>)
 - GUI
 - Easy to use

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Some resources to get started with WEKA

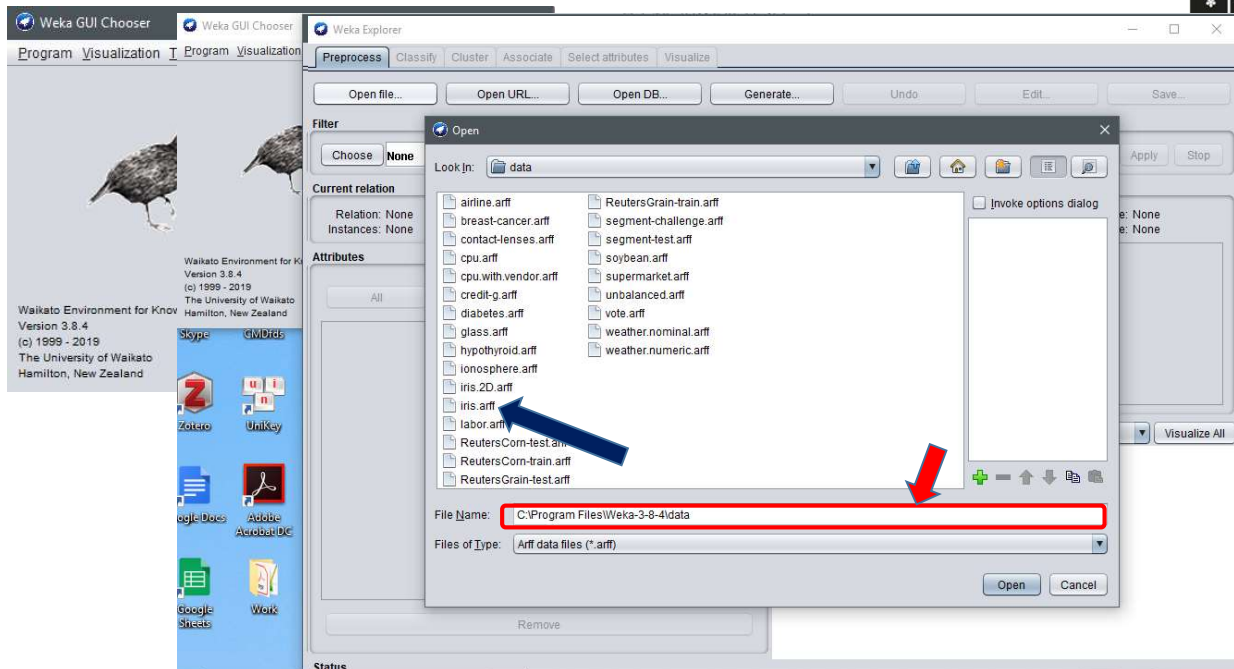


<https://www.analyticsvidhya.com/learning-paths-data-science-business-analytics-business-intelligence-big-data/weka-gui-learn-machine-learning/>

[https://www.cs.waikato.ac.nz/ml/weka/Witten et al 2016 appendix.pdf](https://www.cs.waikato.ac.nz/ml/weka/Witten_et_al_2016_appendix.pdf)

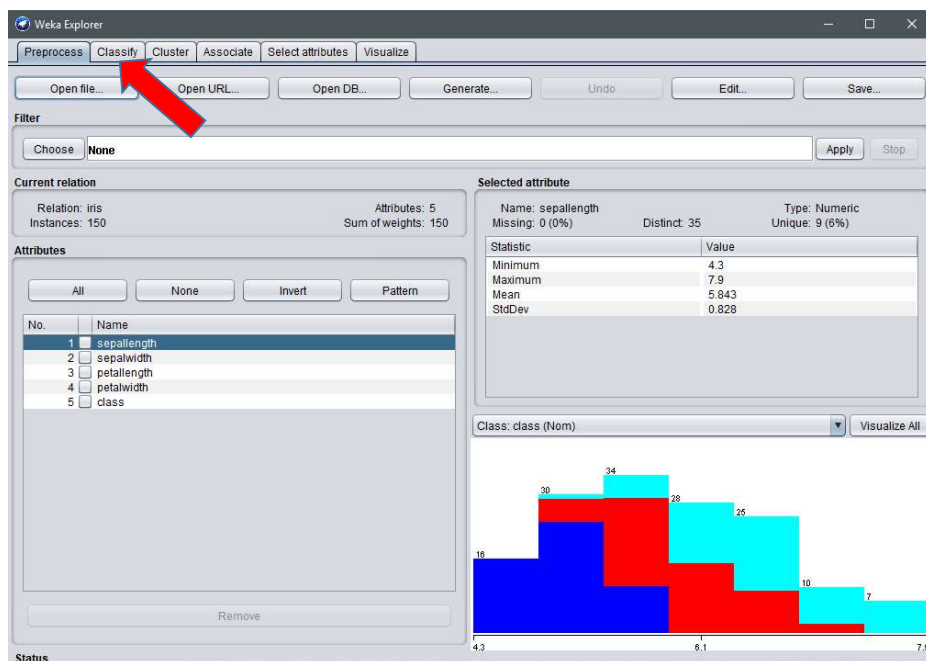
84

WEKA



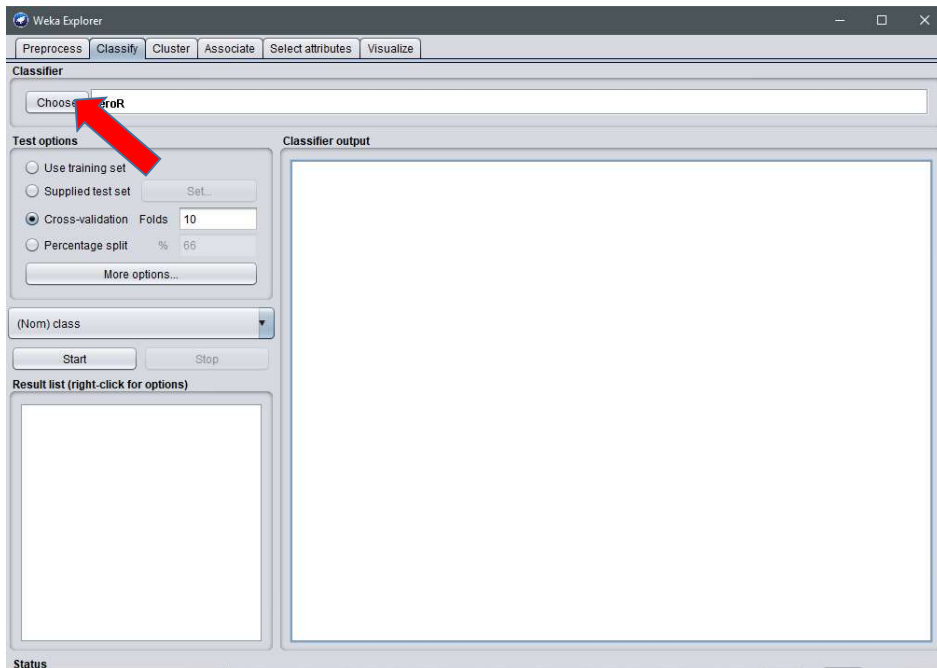
85

WEKA



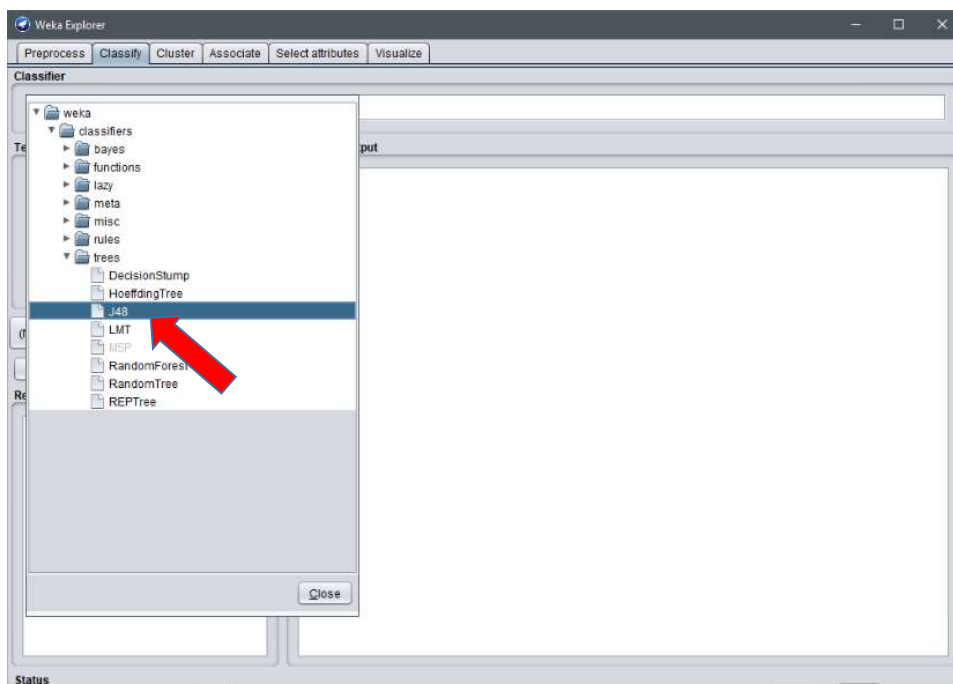
86

WEKA



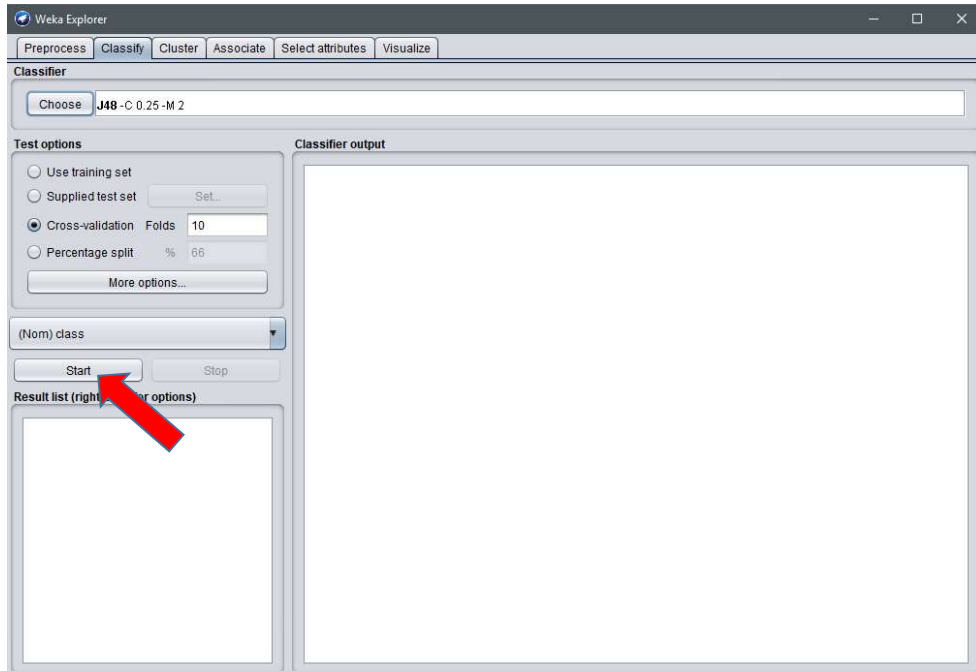
87

WEKA



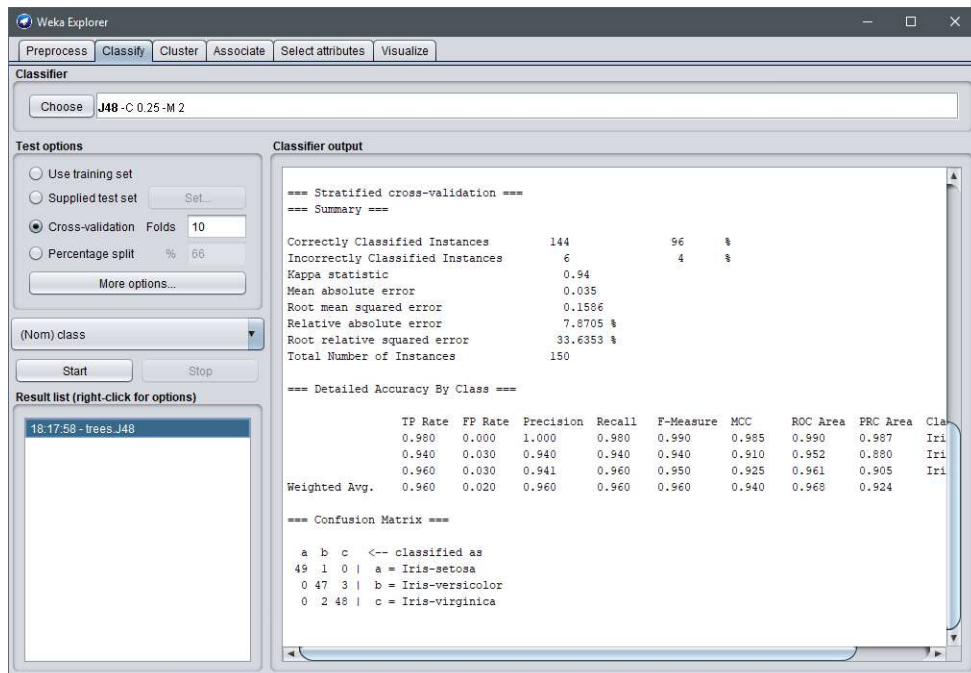
88

WEKA



89

WEKA



90

WEKA

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☐ Supplied test set **Set...**

☒ Cross-validation **Folds 10**

☐ Percentage split **% 66**

More options...

(Nom) class

Start **Stop**

Result list (right-click for options)

18:17:58 - trees.J48

right-click

Classifier output

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 144 96 %

Incorrectly Classified Instances 6 4 %

Kappa statistic 0.94

Mean absolute error 0.035

Root mean squared error 0.1586

Relative absolute error 7.8705 %

Root relative squared error 33.6353 %

Total Number of Instances 150

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.980	0.000	1.000	0.980	0.990	0.985	0.990	0.987	Iris-setosa	
0.940	0.030	0.940	0.940	0.940	0.910	0.952	0.880	Iris-versicolor	
0.960	0.030	0.941	0.960	0.950	0.925	0.961	0.905	Iris-virginica	
0.960	0.020	0.960	0.960	0.960	0.940	0.968	0.924		

Matrix

-- classified as

a = Iris-setosa

b = Iris-versicolor

c = Iris-virginica

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WEKA

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☐ Supplied test set **Set...**

☒ Cross-validation **Folds 10**

☐ Percentage split **% 66**

More options...

(Nom) class

Start **Stop**

Result list (right-click for options)

18:17:58 - trees.J48

Visualize tree

Classifier output

=== Stratified cross-validation ===

=== Summary ===

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0.940	0.030	0.940	0.940	0.940	0.910	0.952	0.880	Iris-versicolor	
0.960	0.030	0.941	0.960	0.950	0.925	0.961	0.905	Iris-virginica	
0.960	0.020	0.960	0.960	0.960	0.940	0.968	0.924		

Matrix

-- classified as

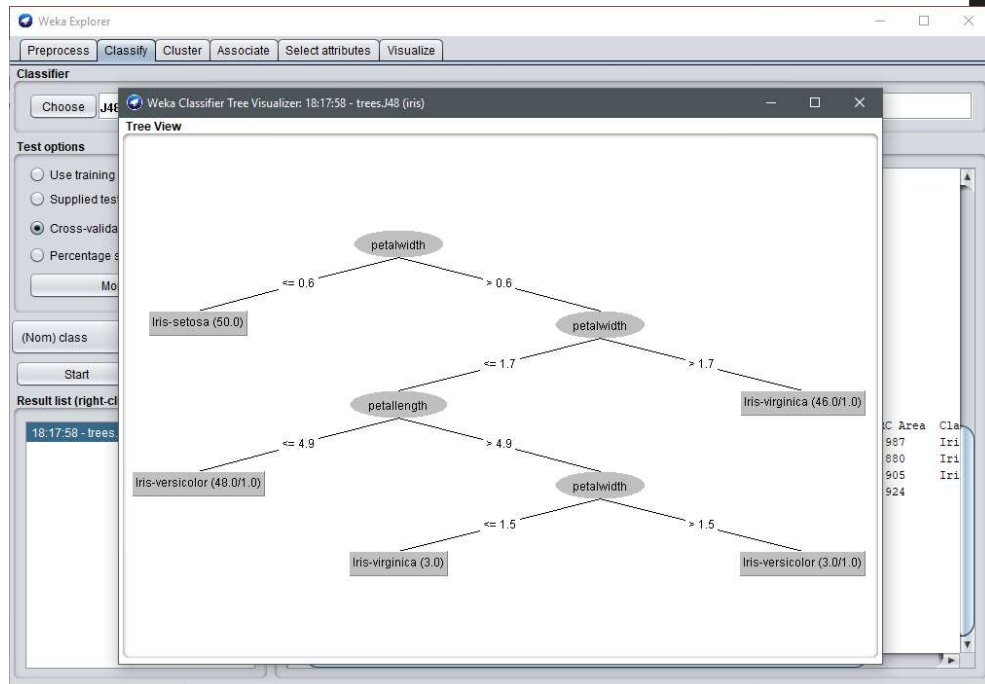
a = Iris-setosa

b = Iris-versicolor

c = Iris-virginica

92

WEKA



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ML for a practitioner

- If you:
 - Are really serious about ML
 - Are ready to spend substantial amount of time to learn
 - Datasets can be really complex/noisy
- Then it's time to move on to a more serious tools:
 - R, python
 - Needs programming, application integration, libraries (tensorflow, pytorch, numpy, scikit-learn, etc.)!

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